

NeetCode 150 — Pattern Recognition Cheat Sheet

18 Patterns • Trigger Signals • Complexity Hints • 5-Min Daily Review

1. Arrays & Hashing

HashMap First

DUPLICATE → Set
COMPLEMENT → Map
FREQUENCY → Counter

Duplicates, pairs, grouping,
 $O(n)$ required

$O(n)$ time, $O(n)$ space

2. Two Pointers

Opposite Ends

SORTED → L/R pointers
PALINDROME → Meet
middle
3SUM → Fix + 2ptr

Sorted array, pairs, partition

$O(n)$ from $O(n^2)$

3. Sliding Window

Expand + Shrink

SUBSTRING → Variable
window
FIXED K → Slide 1 in/out
MIN/MAX → Update
when valid

Longest/shortest contiguous
subarray

$O(n)$ scan once

4. Stack

Monotonic Magic

NEXT GREATER → Pop
when bigger
BRACKETS → Push/pop
match
HISTOGRAM → Width ×
height

Next element, parentheses,
areas

$O(n)$ amortized

5. Binary Search

Answer Space

SORTED → Classic
ROTATED → Half-sorted
check
MIN/MAX CAPACITY →
Feasible(x)

Sorted, $O(\log n)$, minimize
max

$O(\log n)$

6. Linked List

Fast & Slow

CYCLE → Floyd 2x/1x
REVERSE → 3 pointers
MIDDLE → Fast at end

Cycle, middle, reversal, k-
group

$O(n)$ one pass

7. Tree DFS

Global Side-Effect

DEPTH → Return 1+max
DIAMETER → Update
global
BST → Pass (min,max)

Paths, depth, validation, LCA

$O(n)$ visit once

8. Tree BFS

Level Snapshot

LEVEL ORDER → len(q)
snapshot
ZIGZAG → Reverse odd
RIGHT VIEW → Last of
level

Level-by-level, shortest tree
path

$O(n)$

9. Tries

Prefix Tree

PREFIX → children{
WILDCARD → DFS all
BOARD → Trie +
backtrack

Autocomplete, dictionary,
word search

$O(L)$ per word

10. Heap

K-Largest Paradox

K LARGEST → Min-heap
size K
MEDIAN → Two heaps
MERGE K → Heap of
heads

Top K, Kth, streaming
median

$O(n \log k)$

11. Backtracking

Choose + Undo

SUBSETS →
Include/exclude
PERMS → Swap or
used[]
path.pop() → THE undo
step

All combinations, N-Queens,
boards

$O(2^n)$ / $O(n!)$

12. Graphs

DFS/BFS/UF/Topo

ISLANDS → Flood fill
PREREQS → Kahn's topo
CONNECTED → Union-
Find

Grids, connectivity,
dependencies

$O(V+E)$

13. Advanced Graphs

Weighted Shortest

WEIGHTED → Dijkstra
PQ
MST → Prim's/Kruskal
K STOPS → Bellman-
Ford

14. 1-D DP

Take or Skip

STAIRS → Fib
ROBBER → max(skip,
take+prev2)
LIS → Patience $O(n \log n)$

15. 2-D DP

Grid or Two Strings

PATHS → dp[i][j] = sum
LCS → Match or skip
KNAPSACK →
DESCENDING loop

16. Greedy

Local → Global

KADANE → cur=max(x,
cur+x)
JUMP → Track
max_reach
INTERVALS → Sort by
end

17. Intervals

Sort by Start

MERGE → Sort + linear
MIN ROOMS → Two
pointers
NON-OVERLAP → Sort
by end

18. Bit + Math

XOR Cancels

SINGLE → $a^a=0$
COUNT BITS → $n \& (n-1)$
POW → Fast exp x^2

Weighted edges, shortest path, MST

$O(E \log V)$

Count ways, min/max cost, choices

$O(n)$ linear scan

Grids, edit distance, LCS, knapsack

$O(m \cdot n)$

Max/min with local choice

$O(n)$ one pass

Overlapping meetings, merge

$O(n \log n)$

Single number, bit count, rotate

$O(1)$ / $O(\log n)$



NeetCode #1–150 mapped 1:1 | 🧠 5-min daily flashcard drill | 🔥 Print + pin above desk